

The Gaussian (Normal) Distribution 🛎️

To perform anomaly detection, we need a way to calculate the probability $p(x)$ that a specific data point belongs to a dataset. We do this using the **Gaussian Distribution**, also known as the **Normal Distribution** or the **Bell Curve**.

Key Parameters

The shape of the Gaussian distribution is determined by two parameters:

- 1. **μ (Mu) - The Mean:**
 - This describes the **center** of the distribution.
 - It determines where the peak of the bell curve sits.
- 2. **σ (Sigma) - The Standard Deviation:**
 - This describes the **width** of the distribution.
 - It determines how spread out the data points are.
 - σ^2 (**Sigma squared**) is called the **Variance**.

Visualizing the Distribution

The area under the curve always sums to **1** (representing total probability). Therefore, changing the width (σ) forces the height to change as well.

Scenario	Shape of Curve	Interpretation
Standard ($\mu = 0, \sigma = 1$)	Standard Bell Shape	Centered at 0, standard spread.
Small σ (e.g., $\sigma = 0.5$)	Tall and Skinny	Data is tightly clustered around the mean.
Large σ (e.g., $\sigma = 2$)	Short and Wide	Data is very spread out (high variance).
Change μ	Shifts Left/Right	The shape stays the same, but the center moves.

The Mathematical Formula

The probability of x , denoted as $p(x)$, is calculated using the following formula:

$$p(x) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left(-\frac{(x-\mu)^2}{2\sigma^2}\right)$$

- This formula produces the bell-shaped curve.
 - Points near the mean μ will produce a high value for $p(x)$.
 - Points far from the mean (in the "tails") will produce a very low value for $p(x)$.
-

Parameter Estimation: Learning from Data

If you have a dataset of m examples $(x^{(1)}, \dots, x^{(m)})$, how do you find the correct μ and σ to fit your data?

You calculate the **Maximum Likelihood Estimates**:

1. Estimate the Mean (μ):

Simply calculate the average of your training examples.

$$\mu = \frac{1}{m} \sum_{i=1}^m x^{(i)}$$

2. Estimate the Variance (σ^2):

Calculate the average of the squared differences between each example and the mean.

$$\sigma^2 = \frac{1}{m} \sum_{i=1}^m (x^{(i)} - \mu)^2$$

***Note on Statistics:** Some statistics courses recommend dividing by $m - 1$ instead of m for the variance. In machine learning (and for large datasets), the difference is negligible. Using $1/m$ is standard practice here.*

Applying to Anomaly Detection

Once you have calculated μ and σ^2 for your data:

- **Normal examples:** Will fall near the center (μ). $p(x)$ will be **high**.
- **Anomalous examples:** Will fall far away in the tails. $p(x)$ will be **low**.

Currently, we have looked at x as a single number (one feature). However, real-world problems usually involve multiple features $(x_1, x_2, \dots, x_n)$.